

Assignment #1

Course: DSA

CLO	CLO STATEMENT	Bloom's Taxonomy Level	PLO
1	Understand the basic knowledge of data structures and algorithms.	C2	2

Question 1

During a natural disaster, drones are deployed to deliver medical supplies. Each drone has a **priority level** (higher number = more urgent). However, due to limited bandwidth, drones are processed using a **stack** structure, **but only the highest priority drone should be dispatched first**. You are required to design and explain the solution.

Question 2

A robot continuously receives commands in the form of strings (e.g., "MOVE 10", "TURN LEFT"). The buffer starts with a size of 4 and dynamically resizes as more commands are received. Also, oldest commands must be discarded if memory usage exceeds 10 commands.

Question 3

You're helping to build a **web development helper tool**. This tool **reads a simple HTML file line by line** and checks whether **all opening tags have a matching closing tag**.

If any **tag is not closed**, or a **closing tag appears without an opening tag**, the tool will notify the user along with the **line number** of the error.

Note: you need to work only on pair tags.

Deliverables:

- Written explanation of each concept (1–2 paragraphs per task)
- Annotated C++ code or pseudocode
- Optional diagrams for class hierarchy or flow